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Lecture 24 3/23/26

Summary of Last Week:

- Driven Damped oscillations

EOM: $\underbrace{\ddot{x}}_{\frac{ma}{m}} + \underbrace{2\beta\dot{x}}_{-F_{drag}/m} + \underbrace{\omega_0^2 x}_{-F_{spring}/m} = \underbrace{f(t)}_{F_{driving}/m}$

$f(t) \equiv \frac{F(t)}{m}$ $\quad 2\beta \equiv b/m$ $\quad \omega_0 \equiv k/m$

- Linear differential operators: $D(c_1x_1 + c_2x_2) = c_1 Dx_1 + c_2 Dx_2$

- For oscillators: $D = \frac{d^2}{dt^2} + 2\beta\frac{d}{dt} + \omega_0^2$

$Dx = f(t) \Rightarrow$ driven-damped osc. $\Rightarrow$ inhomogeneous

$Dx = 0 \Rightarrow$ damped os

$Dx\Big|_{\beta=0} = 0 \Rightarrow$ undamped, SHO

$\Big\}$ homogeneous

- To solve $Dx = f(t)$

(1) Find some <u>particular</u> solution $Dx_p = f$

(2) Find the <u>homogeneous</u> solution $Dx_H = 0$

(3) Realize $D(x_p + x_H) = Dx_p + Dx_H = f + 0 = f$

(4) General solution $x_g = x_p + \underbrace{C_1 x_H}_{\text{contains two arbitrary constants}}$

- Sinusoida driving force $f(t) = f_0 \cos\omega t$

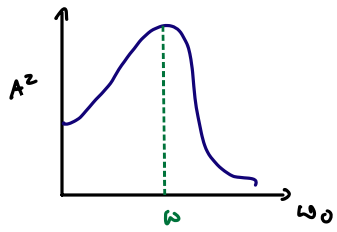
$x_p = A\cos(\omega t + \delta)$ where $A = \frac{f_0^2}{(\omega_0^2-\omega^2)^2 + 4\beta^2\omega^2}$ $\quad \delta = \tan^{-1}\left(\frac{2\beta\omega}{\omega_0^2-\omega^2}\right)$

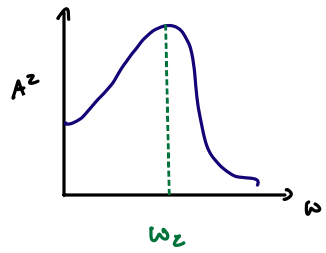
driving ω, not $\omega_0 = \sqrt{k/m}$

$x_H = C_1 e^{r_1 t} + C_2 e^{r_2 t}$

$x_g = A e^{-\beta t}\cos(\omega t - \delta) + C_1 e^{r_1 t} + C_2 e^{r_2 t}$

- Resonance





$\omega_2 = \sqrt{\omega_0^2 - 2\beta^2} \approx \omega_0$ for small β

Find by minimizing denominator of A

Important frequencies: $\omega_0 = \sqrt{\frac{k}{m}}$ "natural" $\quad \omega_1 = \sqrt{\omega_0^2 - \beta^2}$ "damped" $\quad \omega$ "driven" $\quad \omega_2 = \sqrt{\omega_0^2 - 2\beta^2}$ "resonance"

- Fourier Series:

Any periodic function $f(t+\tau) = f(t)$ can be rewritten as:

$$f(t) = \sum_{n=0}^{\infty} \left[a_n \cos(n\omega t) + b_n \sin(n\omega t) \right] \qquad \omega = 2\pi/\tau$$

$$a_0 = \frac{1}{\tau} \int_{-\tau/2}^{\tau/2} f(t)\, dt \qquad [n=0]$$

$$a_n = \frac{2}{\tau} \int_{-\tau/2}^{\tau/2} f(t) \cos(n\omega t)\, dt$$

$$b_n = \frac{2}{\tau} \int_{-\tau/2}^{\tau/2} f(t) \sin(n\omega t)\, dt$$

do not just plug $f(t) = \sum_{n=0}^{\infty} a_n \cos n\omega t + b_n \sin n\omega t$ back in!

You need specific info about $f(t)$ to determine a_n, b_n

If $f(t) = \text{odd}$, $a_n = 0$

$f(t) = \text{even}$, $b_n = 0$

Fourier Series for the Driven Oscillator

EOM: $\ddot{x} + 2\beta \dot{x} + \omega_0^2 x = f$

$$Dx = f \quad , \quad D = \frac{d^2}{dt^2} + 2\beta \frac{d}{dt} + \omega_0^2$$

$Dx_1 = f_1 \qquad Dx_2 = f_2 \quad \Rightarrow \quad D(x_1 + x_2) = f_1 + f_2 \qquad$ (linear)

We can go further. $f(t) = \sum_n f_n(t)$ where each f_n gives soln x_n s.t. $Dx_n = f_n$

Let $x = \sum_n x_n(t)$

Then $Dx = D\left(\sum_n x_n(t)\right) = \sum_n Dx_n(t) = \sum_n f_n(t) = f(t)$

$Dx = f \Rightarrow x$ is a solution!

Now let's combine this with Fourier Series.

<u>Any</u> τ-periodic $f(t)$ can be written as $f(t) = \sum_{n=0}^{\infty} a_n \cos(n\omega t) + b_n \sin(n\omega t)$

For simplicity, let's consider even $f(t) \Rightarrow f(t) = \sum_{n=0}^{\infty} f_n \cos(n\omega t)$

Remember, when $f(t) = f_0 \cos\omega t$, $x(t) = A\cos(\omega t - \delta)$

now $f(t) = f_0 \cos n\omega t \Rightarrow x_n(t) = A_n \cos(n\omega t - \delta_n)$

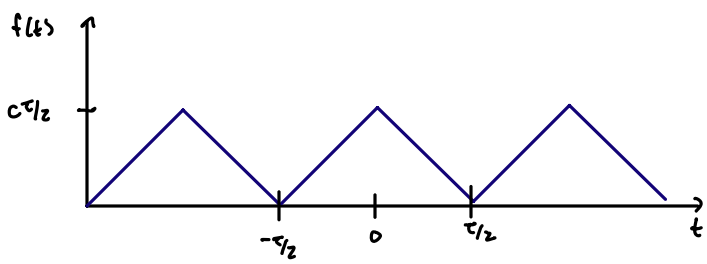
$$A_n^2 = \frac{f_n^2}{(\omega_0^2 - n^2\omega^2)^2 + 4\beta^2 n^2 \omega^2}$$

$$\delta_n = \left(\tan^{-1} \frac{2\beta n\omega}{\omega_0^2 - n^2\omega^2}\right)$$

$\Rightarrow x(t) = \sum_n x_n(t) = \sum_n A_n \cos(n\omega t - \delta_n)$

1) Find f_n
2) Find A_n, $\delta_n \Rightarrow x_n(t)$
3) Write down $x(t) = \sum_n x_n(t)$

Example:



rise $= \pm c\tau/2$
run $= \tau/2$
$\Rightarrow$ slope $= \pm c$

$f(t) = \sum_n a_n \cos n\omega t + b_n \sin n\omega t$

$$a_0 = \frac{1}{\tau}\int_{-\tau/2}^{\tau/2} f(t)\,dt \qquad f(t) = \begin{cases} ct + c\tau/2 & -\tau/2 < t < 0 \\ -ct + c\tau/2 & 0 < t < \tau/2 \end{cases}$$

This function is even $\Rightarrow b_n = 0$

$$= \frac{1}{\tau}\int_{-\tau/2}^{0} (ct + c\tau/2)\,dt + \frac{1}{\tau}\int_{0}^{\tau/2} (-ct + c\tau/2)\,dt$$

$$= \frac{c}{\tau}\left[\frac{1}{2}t^2 + \tau/2\, t\right]_{-\tau/2}^{0} + \frac{c}{\tau}\left[-\frac{1}{2}t^2 + \tau/2\, t\right]_{0}^{\tau/2}$$

$$= \frac{c}{\tau}\left[-\frac{1}{2}\tau^2/4 + \tau/2\,\tau/2\right] + \frac{c}{\tau}\left[-\frac{1}{2}\tau/4 + \tau/2\,\tau/2\right]$$

$$= \frac{2c}{\tau}\left[-\tau^2/8 + \tau^2/4\right] = \frac{2c}{\tau}\left[\tau^2/8\right] = c\tau/4$$

$$a_n = \frac{2}{\tau}\int_{-\tau/2}^{\tau/2} f(t)\cos n\omega t\, dt$$

$$= \frac{2}{\tau}\int_{-\tau/2}^{0} c(t+\tau/2)\cos n\omega t\, dt + \frac{2}{\tau}\int_{0}^{\tau/2} c(-t+\tau/2)\cos n\omega t\, dt$$

$$= \frac{2c}{\tau}\left[\int_{-\tau/2}^{0} t\cos n\omega t + \tau/2\int_{-\tau/2}^{0}\cos n\omega t\, dt - \int_{0}^{\tau/2} t\cos n\omega t + \tau/2\int_{0}^{\tau/2}\cos n\omega t\, dt\right]$$

$$\int x\cos ax = \frac{1}{a^2}\cos ax + \frac{x}{a}\sin ax$$

$$= \frac{2c}{\tau}\left[\left(\frac{1}{n^2\omega^2}\cos n\omega t + \frac{t}{n\omega}\sin n\omega t\right)_{-\tau/2}^{0} + \tau/2\left(\frac{1}{n\omega}\sin n\omega t\right)_{-\tau/2}^{0} - \left(\frac{1}{n^2\omega^2}\cos n\omega t + \frac{t}{n\omega}\sin n\omega t\right)_{0}^{\tau/2} + \tau/2\left(\frac{1}{n\omega}\sin n\omega t\right)_{0}^{\tau/2}\right]$$

$$= \frac{2c}{\tau}\left[\frac{1}{n^2\omega^2}(1) + 0 - \frac{1}{n^2\omega^2}\cos\left(\frac{n\omega\tau}{2}\right) - \left(\frac{-\tau}{2n\omega}\right)\sin\left(-\frac{n\omega\tau}{2}\right) + 0 - \tau/2\left(\frac{1}{n\omega}\right)\sin\left(-\frac{n\omega\tau}{2}\right) - \left(\frac{1}{n^2\omega^2}\cos\left(\frac{n\omega\tau}{2}\right) + \frac{\tau}{2n\omega}\sin\left(\frac{n\omega\tau}{2}\right) - \frac{1}{n^2\omega^2} - 0\right) + \tau/2\left(\frac{1}{n\omega}\sin\frac{n\omega\tau}{2}\right)\right]$$

$$\omega = \frac{2\pi}{\tau} \qquad \frac{n\omega\tau}{2} = \frac{2\pi}{\tau}\frac{n\tau}{2} = \pi n$$

$$= \frac{2c}{\tau}\left[\frac{1}{n^2\omega^2} - \frac{1}{n^2\omega^2}\cos n\pi - \cancelto{0}{\frac{\tau}{2n\omega}\sin n\pi} + \cancelto{0}{\frac{\tau}{2n\omega}\sin n\pi} - \frac{1}{n^2\omega^2}\cos n\pi - \cancelto{0}{\frac{\tau}{2n\omega}\sin n\pi} + \frac{1}{n^2\omega^2} + \cancelto{0}{\frac{\tau}{2n\omega}\sin n\pi}\right]$$

$$= \frac{2c}{\tau}\left[\frac{2}{n^2\omega^2} - \frac{2}{n^2\omega^2}\cos n\pi\right]$$

$$= \frac{2c}{\tau}\left(\frac{2}{n^2\omega^2}\right)(1-\cos n\pi)$$

$$= \frac{2c}{\tau}\left[\frac{2\tau}{n^2\omega\, 2\pi}\right](1-\cos n\pi)$$

$$= \frac{2c}{\pi n^2\omega}(1-\cos n\pi) \qquad \cos(n\pi) = (-1)^n$$

$$= \frac{2c}{\pi n^2\omega}\left[1-(-1)^n\right]$$

$$a_n = \begin{cases} \dfrac{4c}{\pi n^2\omega} & n = \text{odd} \\ 0 & n = \text{even} \end{cases}$$